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Machine Learning-Assisted End-to-End Synthesis of RF Transformer-Based Matching Networks

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***Abstract*—This paper proposes a learning-driven synthesis framework for RF transformer-based matching networks with three-dimensional electromagnetic (EM) structures. Unlike conventional methods that derive EM structures from target circuit element values and topologies, the proposed method directly synthesizes passive networks from desired performance metrics and given topologies. Using on-chip transformer impedance matching networks as a case study, the model first extracts key features from input impedance and load capacitance via neural networks to predict initial geometric parameters of the transformer's 3D EM structure in 28-nm TSMC CMOS technology. Subsequently, Bayesian optimization (BO) refines these geometric parameters for precise performance tuning. The resulting structures achieve arbitrary impedance matching while simultaneously accommodating two load capacitances. As a proof of concept, multiple transformer designs were synthesized and validated by Cadence simulations, confirming their compliance with the targeted impedance characteristics.**

Keywords—EM optimization, machine learning, inverse design, end-to-end synthesis.

I. Introduction

Passive matching networks are essential in RF systems to ensure maximum power transfer, noise matching, and impedance transformation [1], [2]. Despite the diversity of available topologies, their design still heavily relies on full-wave EM simulations. This makes the process time-consuming, resource-intensive, and dependent on expert experience [3]-[5]. Typically, designers must build and simulate the EM structure before evaluating key metrics such as S-parameters, impedance, loss, and bandwidth. As network complexity grows, simulation time can extend to several hours, and multiple optimization iterations are often required [6]-[8]. An end-to-end synthesis tool that generates EM geometry directly from the network topology and target specifications would significantly streamline this process, reducing design time and manual effort.

Many recent advances rely heavily on high-fidelity surrogate models, which shift the cost from full-wave simulations to large-scale data collection. However, training such models for EM tasks often requires tens of thousands to millions of labeled samples, demanding months or even years of simulation time. This makes them impractical for time- and resource-constrained industrial applications.

To address these challenges, we propose a data-efficient end-to-end synthesis framework that directly generates transformer-based impedance matching networks operating at 28 GHz. Given circuit-level specifications, including the source and load impedances (Z_1, Z_2) and target S-parameters,

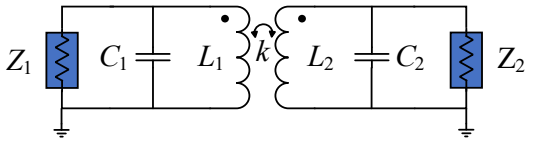


Fig. 1. Schematic of transformer-based impedance matching network.

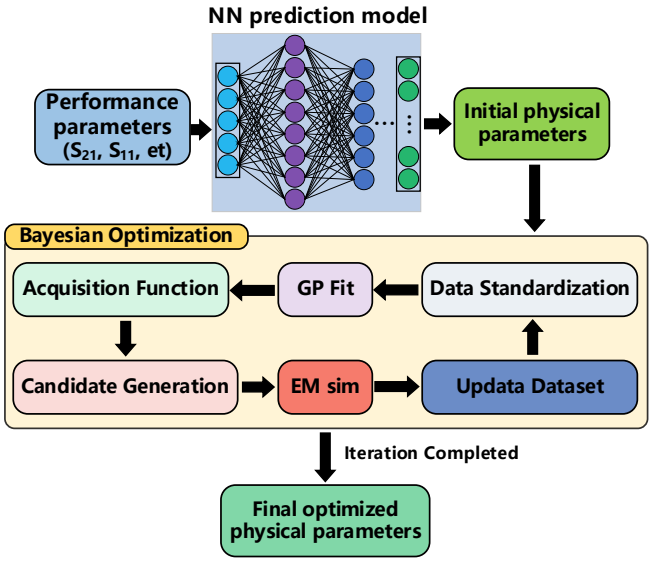


Fig. 2. End-to-end inverse design framework combining neural network and BO.

the framework predicts the EM structure and corresponding shunt capacitors. As illustrated in Fig. 1, the network incorporates non-ideal magnetic coupling and shunt capacitors (C_1, C_2) at both ports. By synthesizing the EM geometry directly from performance targets, the proposed method significantly reduces design iterations and computational overhead compared with conventional EM-driven optimization workflows.

II. End-to-End Synthesis Model

The proposed framework automates the synthesis process through two sequential modules, as illustrated in Fig. 2. A neural network first extracts features from the target impedance and S-parameters to generate a coarse prediction of the transformer geometry and shunt capacitance in a 28 nm CMOS process [9]. This prediction is then refined by a BO module that searches within a constrained design space to further improve performance.

A. *Initial Prediction with Neural Network*

While neural networks (NNs) have demonstrated outstanding performance in various domains, their architectures must be carefully tailored to specific tasks. In this end-to-end synthesis problem, one key challenge lies in predicting a high-dimensional physical parameter set (11 variables) from a low-dimensional input (5 performance metrics). This mapping is inherently ill-posed due to information loss: physical parameters determine performance through deterministic physical laws, but the reverse is not uniquely defined. According to the data processing inequality, performance metrics may contain less information than the full set of physical variables, making inverse prediction fundamentally difficult. Therefore, the neural network is employed to provide a coarse estimate of the transformer geometry and shunt capacitance based on a 28-nm TSMC CMOS process. Fig. 3 illustrates the synthesized transformer structure along with the input and output parameters of the neural network.

A total of 22k samples were generated using Cadence EM simulations. After shuffling, 20k samples were randomly selected for training and the remaining 2k were reserved for testing. To mitigate overfitting, weight regularization was applied to all trainable parameters. The input and output layers consist of 5 and 11 elements, corresponding to the target circuit performance parameters and the predicted transformer design parameters including the shunt capacitances, respectively. As shown in Fig. 4, by properly selecting the network depth, a robust predictive model can be constructed. Experimental results indicate that a 6-layer fully connected network achieves lower absolute errors and a smaller gap in RMSE between training and testing datasets, suggesting improved generalization and reduced overfitting compared to shallower or deeper architectures.

B. *Refinement with Bayesian Optimization*

To further enhance the accuracy of transformer-based impedance matching network synthesis, a second-stage refinement process is applied to the geometry parameters predicted by the neural network. Specifically, the predicted transformer three-dimensional EM structure and the associated capacitance values are used as the initialization for a constrained optimization procedure. The neural network achieves an average Mean Squared Error (MSE) of approximately 0.7 on the test set, corresponding to an average per-parameter error of roughly 0.06 across the 11 design variables. Based on this analysis, a ±20% search range is defined around each predicted parameter to construct a feasible design space for further optimization. This range is sufficiently wide to include potentially better-performing configurations, while remaining close enough to the initial estimates to maintain physical feasibility and ensure stable convergence during the optimization process.

To efficiently explore this design space, BO is adopted as the refinement strategy. BO is a sample-efficient global optimization method that is particularly well suited for problems where objective function evaluations are expensive, such as full-wave EM simulations. In this framework, a Gaussian Process (GP) surrogate model is first constructed to approximate the relationship between the design parameters and the circuit performance. An acquisition function, specifically the Expected Improvement (EI), is then employed to balance exploration and exploitation when selecting new candidate solutions. These candidates are subsequently evaluated using EM simulations, and the resulting data are iteratively incorporated into the surrogate model to progressively improve its accuracy. The optimization objective is to minimize a performance metric that jointly reflects transmission efficiency and impedance matching characteristics, thereby guiding the search toward high-performance transformer-based matching network designs.

To evaluate the performance of BO, we compare it against several commonly used optimization algorithms, including Random Search and Genetic Algorithm (GA). All algorithms are initialized from the same set of starting samples and are allowed to run for the same number of iterations to ensure fairness in comparison. To mitigate the effects of randomness inherent in stochastic optimization methods, each algorithm is independently repeated ten times. Table I presents the comparative results among the three methods. The

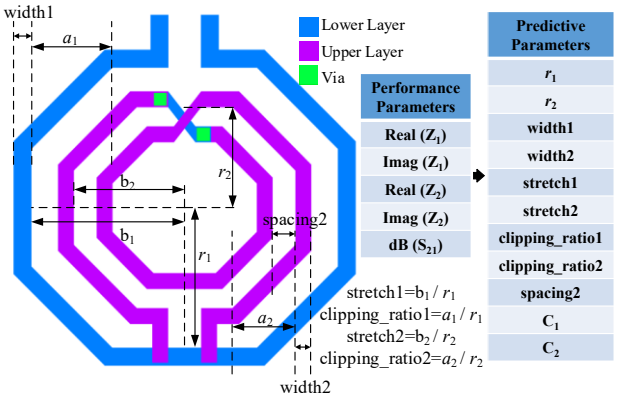


Fig. 3. Synthesized transformer structure and neural network input-output parameters.

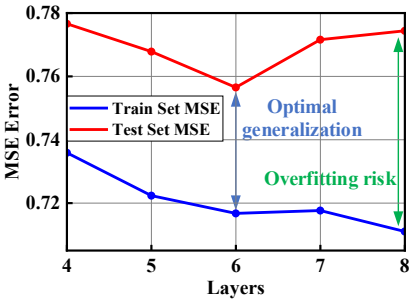


Fig. 4. RMSE versus number of fully connected layers for overfitting prevention.

TABLE I. PERFORMANCE COMPARISON OF OPTIMIZATION ALGORITHMS. (10 RUNS)

Method	Best S_{21} (dB)	Std. Dev. of Best S_{21}	Top-10 S_{21} Mean (dB)	Std. Dev. of All S_{21}
Random Search	-1.29	0.1815	-2.5086	6.4390
Genetic Algorithm	-1.30	0.2349	-1.9868	**6.0234**
BO(Ours)	**-1.13**	**0.1385**	**-1.8383**	6.1178

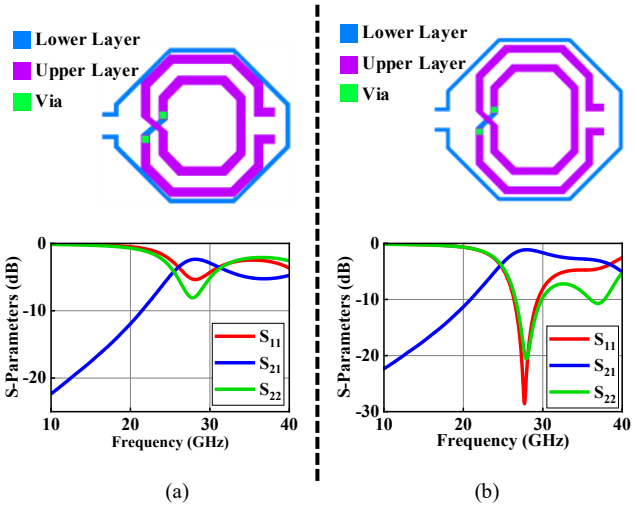


Fig. 5. (a) Neural network prediction and (b) BO refinement of transformer geometry and S-parameters.

performance is evaluated using four metrics: the best S_{21} value obtained across all iterations, the standard deviation of the best S_{21} over the ten repeated runs to reflect consistency, the average S_{21} value of the top 10 candidates in each run as an indicator of overall solution quality, and the standard deviation of all S_{21} values produced during optimization to characterize the exploration behavior of each algorithm. As shown in the table, BO achieves the best transmission performance with the lowest variability, demonstrating superior accuracy and robustness compared to Random Search and GA. This underscores the advantage of combining deep learning-based initialization with probabilistic optimization, resulting in a practical and scalable framework for RF passive component synthesis.

III. Matching Network Design and Verification

The inverse design of matching networks essentially enables end-to-end circuit synthesis through algorithmic automation, eliminating the need for manual tuning during the optimization process. To demonstrate the potential of the transformer-based design methodology, we evaluate the entire optimization process. First, specific port impedances—Z_1 for Port 1 and Z_2 for Port 2—as well as S-parameter targets are fed into a neural network prediction model. The model provides a coarse estimation of the transformer's 3D EM structural parameters along with the associated parallel capacitances. The resulting predicted transformer geometry and corresponding S-parameters of the matching network are shown in Fig. 5(a). Next, based on the predicted geometrical parameters and capacitance values, we define a ±20% search range and apply BO to iteratively refine the parameters. As shown in Fig. 5(b), the performance of the matching network is significantly improved after optimization—for example, the S_{21} is enhanced from -2.4 dB to -1.3 dB. This integrated framework enables direct synthesis of transformer geometry and associated capacitance values, achieving end-to-end synthesis of transformer-based impedance matching networks.

IV. Conclusion

In this work, we proposed a machine learning-assisted framework for the inverse synthesis of transformer-based impedance matching networks operating at frequencies. The proposed framework integrates a neural network for coarse prediction of 3D transformer geometry and parasitic capacitance based on port impedance and desired S-parameters. This is followed by BO within a constrained design range to refine the geometric parameters and enhance matching performance. Experimental results demonstrate that the synthesized structures can achieve precise impedance matching, with the final S_{21} performance significantly improved. Validated through EM simulations on a 28nm CMOS process, our inverse design methodology reduces the need for manual tuning and significantly accelerates the design cycle. This study highlights the potential of data-driven techniques in synthesizing RF passive structures, paving the way for efficient, automated circuit-level design solutions.

V. Acknowledgement

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